

CUET PG MCA (DIGITAL LOGIC PYQS)

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Topics Covered: Logic Gates, Combinational Circuits (Half/Full Adder, Mux, Decoder, Encoder, Demux), Sequential Circuits (Flip-Flops, Latches, Registers), Memory Organization.

Question 1

CUET PG 2025 | Logic Gate

The following circuit generates the same output as?

- | | |
|---------------|---------------|
| (a) XOR GATE | (b) XNOR GATE |
| (c) NAND GATE | (d) NOR GATE |

Question 2

CUET PG 2025 | Multiplexer

Minimum number of 2:1 Multiplexers required to design a 16:1 Multiplexer is?

- | | |
|--------|--------|
| (a) 12 | (b) 14 |
| (c) 15 | (d) 16 |

Question 3

CUET PG 2025 | Sequential Circuits

Consider the following statements.

- (A) Combinational logic circuits do not have memory, while sequential logic circuits have memory elements.
- (B) Sequential logic circuits depend only on the current inputs, while combinational logic circuits depend on both current and past inputs.
- (C) Flip-flops and latches are examples of combinational logic circuits.
- (D) Multiplexer is an example of combinational logic circuit.

Choose the correct statement/statements:

- | | |
|--------------------------------|---------------------------|
| (a) (A) and (D) only | (b) (A), (B) and (D) only |
| (c) (A), (B), (C) and (D) only | (d) (B), (C) and (D) only |

Question 4

CUET PG 2025 | Decoder

Arrange the following steps in the correct order to understand the functioning of a 3 to 8 line decoder.

- (A) The decoder enable inputs is set to 1.
- (B) The decoder activates one of its 8 output lines based on the input code.
- (C) The input binary code is applied to the decoder.
- (D) The decoder converts the 3-bit binary input into 8 possible outputs.

- | | |
|------------------------|------------------------|
| (a) (A), (B), (C), (D) | (b) (A), (D), (C), (B) |
| (c) (A), (C), (D), (B) | (d) (A), (B), (D), (C) |

Question 5

CUET PG 2025 | JK Flip Flop

Match List-I with List-II

List-I (JK Flip Flop inputs): (a) 0 0 (b) 0 1 (c) 1 0 (d) 1 1

List-II (Next State): (i) Toggle (ii) Set (iii) Reset (iv) No Change

- | | |
|--|--|
| (a) (A)–(i), (B)–(ii), (C)–(iii), (D)–(iv) | (b) (A)–(iv), (B)–(iii), (C)–(ii), (D)–(i) |
| (c) (A)–(iv), (B)–(iii), (C)–(i), (D)–(ii) | (d) (A)–(i), (B)–(ii), (C)–(iv), (D)–(iii) |

Question 6

CUET PG 2025 | Memory Organization

To provide a memory capacity of $32K \times 16$ how many address lines and data lines are required?

- | | |
|-------------------------------------|-------------------------------------|
| (a) address lines=14, data lines=16 | (b) address lines=15, data lines=16 |
| (c) address lines=14, data lines=4 | (d) address lines=15, data lines=4 |

Question 7

CUET PG 2026 | Half Adder

A half adder has:

- | | |
|---------------------------------|--------------------------------|
| (1) One input and two outputs | (2) Two inputs and two outputs |
| (3) Three inputs and one output | (4) Two inputs and one output |

Question 8

CUET PG 2026 | JK Flip Flop

A JK flip flop behaves as a T flip flop when:

- | | |
|--------------------|--------------------|
| (1) $J = 0, K = 1$ | (2) $J = 1, K = 0$ |
| (3) $J = 1, K = 1$ | (4) $J = 0, K = 0$ |

Question 9

CUET PG 2026 | Combinational Circuits Match

Match List-I with List-II

List-I: (A) Decoder (B) Multiplexer (C) Encoder (D) Demultiplexer

List-II: (I) Selects one input out of many (II) Converts binary input into 2^n outputs (III) Converts 2^n inputs into binary code (IV) Routes one input to several outputs

- | | |
|----------------------------|----------------------------|
| (1) A-I, B-II, C-III, D-IV | (2) A-II, B-I, C-III, D-IV |
| (3) A-III, B-II, C-IV, D-I | (4) A-IV, B-III, C-II, D-I |

Question 10

CUET PG 2026 | Registers

The fundamental function of a register in digital circuits is to:

- | | |
|-----------------------------------|------------------------------|
| (1) Perform Arithmetic operations | (2) Store Binary information |
| (3) Compare two numbers | (4) Encode input signals |

Question 11

CUET PG 2026 | Flip Flops Assertion

Assertion (A): Flip flops are used in counters and registers.

Reason (R): Flip flops are combinational circuits that store data temporarily.

- | |
|--|
| (1) Both (A) and (R) are correct and (R) is the correct explanation of (A) |
| (2) Both (A) and (R) are correct but (R) is not the correct explanation of (A) |
| (3) (A) is correct but (R) is not correct |
| (4) (A) is not correct but (R) is correct |

Question 12

CUET PG 2026 | Combinational vs Sequential

Which of the following are Combinational Circuits?

- (A) Half adder
- (B) SR flip-flop
- (C) Full adder
- (D) D flip-flop

(1) A and C only

(2) B and D only

(3) B, C, D only

(4) A, B, C, D

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ANSWER KEY (2021-2026)

Q.No	Ans	Q.No	Ans	Q.No	Ans
1	b	2	d	3	a
4	c	5	b	6	b

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